

Recollection Rejection: How Children Edit Their False Memories

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A new methodology is presented for studying children's ability to suppress memory reports of false-but-gist-consistent events, one that measures children's use of a specific editing operation (recollection rejection) that suppresses false reports by accessing verbatim traces of true events. Children make memory reports under 2 instructional conditions, verbatim and gist, and the data are analyzed with fuzzy-trace theory's conjoint-recognition model. Application of the new methodology in studies of children's false memory for narrative events revealed that (a) false-memory editing increases dramatically between early and middle childhood, (b) even young children spontaneously edit their false memories, (c) measures of children's false-memory editing react appropriately to experimental manipulations, and (d) developmental reductions in the incidence of false-memory reports are primarily due to developmental improvements in verbatim memory ability (rather than to decreases in the formation of false memories). Implications for child forensic interviewing are discussed.

During the previous decade, the ways in which children's recollections are influenced by memory suggestion were extensively studied (for reviews, see Bruck & Ceci, 1997; Ceci & Bruck, 1993, 1995). A core finding was that children's memories can be falsified by suggestions ranging from the blatant and deliberate to the subtle and inadvertent (Reyna & Lloyd, 1997). Because suggestive questioning is standard practice in police investigative interviews (see Inbau, Reid, & Buckley, 1986), this finding fomented much concern about the tainting of children's memory reports in certain types of legal cases (e.g., *State v. Michaels*, 1994). This concern, in turn, encouraged the development of research-based interviewing protocols for use with children (Lamb, Orbach, Sternberg, Hershkowitz, & Horowitz, 2000; Lamb, Sternberg, Orbach, Hershkowitz, & Esplin, 1999; Poole & Lamb, 1998). Such protocols restrict suggestive questioning, with a view toward reducing the tendency of forensic interviews to implant false memories.

Despite the existence of such protocols, false-memory reports are a continuing concern in child forensic interviewing, for at least three reasons. First, interviewers who are trained in the use of these protocols routinely depart from them in the field, resorting to suggestive lines of questioning (Myklebust & Alison, 2000; Warren, Woodall, Hunt, & Perry, 1996). Second, even when nonsuggestive protocols are scrupulously followed during formal interviews, children may be exposed to memory suggestion outside the interview context (e.g., in

conversations with parents or other relatives). Third, even in the unlikely event that it were possible to eliminate suggestion from both formal interviews and informal conversations, research on spontaneous memory falsification shows that the contents of children's memories will nevertheless be falsified, to some degree, by endogenous distortion mechanisms that are intrinsic to human memory (for reviews, see Brainerd & Reyna, 1998a; Reyna & Brainerd, 1995). Those mechanisms include autosuggestions that are by-products of thinking about events (Brainerd & Reyna, 1993, 1995; Reyna, 1995) and erroneous inferences that fill in verbatim details of experience that have been lost with the passage of time (Ornstein, Shapiro, Clubb, Follmer, & Baker-Ward, 1997; Pipe, Gee, Wilson, & Egerton, 1999; Pipe & Wilson, 1994; Poole & White, 1993; Reyna & Kiernan, 1994).

Because the formation of at least some false memories is inevitable, additional interviewing tools are required beyond simply reducing the incidence of suggestive questioning. A possibility that has provoked much interest is that children may possess *cognitive editing operations* that allow them to suppress the reporting of certain types of false memories (Brainerd & Reyna, 1998a; Koriati, Goldsmith, Schneider, & Nakash-Dura, 2001). If children possess such editing operations, interviewing procedures could be devised that encourage their use. In line with this possibility, there are prior studies in which certain memory-test manipulations that reduce children's false-memory reports (e.g., cuing, instructions) have been interpreted as showing that children sometimes suppress such reports by, for instance, making metacognitive judgments about how memories originated (e.g., Lindsay, Gonzales, & Eso, 1995) or monitoring memory strength (e.g., Koriati et al., 2001). As yet, however, little progress has been made on this matter because, unlike memory suggestion, false-memory editing lacks the necessary basic research foundation. In particular, we lack clear theoretical definitions of specific editing operations, and we lack unambiguous experimental demonstrations that children use those operations.

Our aim in the present article is to make progress on both of these fronts. With respect to the first, fuzzy-trace theory provides a detailed process model of a candidate editing operation, called *recollection rejection*, that should be within the cognitive capabilities of even young children (Brainerd, Reyna, & Mojardin, 1999; Brainerd, Wright, Reyna, & Mojardin, 2001). We define and

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This research was supported in part by National Science Foundation Grant SBR-9730143, by National Institutes of Health Grants NIH31620 and P50AT00008, and by a Schering Plough/Integrated Therapeutics Group grant.

We thank Morris Goldsmith for his comments on an earlier version of this article.

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illustrate that editing operation in the first section below. On the second front, we introduce a new paradigm, called *release from recollection rejection*, with two important features. The first is that children respond to memory probes under two types of instructional conditions: what *did* happen versus what *could have* happened given the gist of experience. The second feature is a simple mathematical model that uses performance in both conditions to extract direct measurements of recollection rejection. We apply the release-from-recollection-rejection paradigm to children's false memory for narratives using three sets of data. Among the major findings are clear evidence of recollection rejection in young children and in language-disabled children, large developmental improvements in recollection rejection, and evidence that the editing operation responded to experimental manipulations in ways that were consistent with its theoretical definition.

Fuzzy-Trace Theory and Recollection Rejection

Although there are other theoretical approaches to false memory, notably constructivism (e.g., Bransford & Franks, 1971) and the source monitoring framework (e.g., Johnson, Hashtroudi, & Lindsay, 1993), we rely on fuzzy-trace theory in this article. One reason for this choice is that fuzzy-trace theory has been successful in explaining several key findings about false memory that seem to violate other approaches (for a review, see Reyna & Lloyd, 1997). A second reason is that fuzzy-trace theory has generated a series of counterintuitive predictions about false memory that have been confirmed in experimentation, such as the false-memory persistence effect (Brainerd, Reyna, & Brandse, 1995; Payne, Elie, Blackwell, & Neuschatz, 1996), the mere-testing effect (Brainerd & Mojardin, 1998; Brainerd & Reyna, 1996), the false-recognition reversal effect (Brainerd, Reyna, & Kneer, 1995; Brainerd et al., 1999), and the phantom recollection effect (Brainerd, Wright, Reyna, & Mojardin, 2001; Payne et al., 1996). A third reason is that fuzzy-trace theory provides a candidate false-memory editing operation that satisfies both of the above criteria. That is, the posited operation is grounded in a detailed process theory that also specifies a paradigm that delivers direct measurements of the operation.

Fuzzy-trace theory's analysis of children's false memories (Brainerd & Reyna, 1998a, 2001) consists of principles that specify the types of representations that are initially stored and how those representations are subsequently retrieved on memory tests. With respect to storage, a variety of evidence suggests that as children encode events (e.g., words on a list, statements in a narrative, occurrences in everyday life), they store two types of representations: those that capture events' exact surface forms (verbatim traces) and those that capture semantic concepts and other relations that the events exemplify (gist traces). Studies of the time course of storage suggest that verbatim and gist traces are deposited roughly in parallel, rather than gist traces being extracted after verbatim storage is complete. Specifically, event perception initiates storage of surface information and, simultaneously, storage of concepts and relations that are represented in long-term memory but are given local episodic interpretations (Brainerd & Reyna, 1993). Parallel storage means that verbatim and gist traces are *dissociated* representations of experience, which is crucial in explaining how true and false memories of the same experiences can coexist (Brainerd & Reyna, 2001; Reyna, 2001). In studies of memory suggestion, parallel storage of verbatim and gist traces

occurs when original events are experienced and also when memory suggestions are experienced (Reyna, 1995).

With regard to retrieval, the standard measures of false memory are (a) false acceptance of nonexperienced events on recognition tests and (b) intrusions of nonexperienced events on recall tests. In studies of spontaneous false memory, events that are falsely recognized or recalled are overwhelmingly ones that preserve the gist of experience, as when a child falsely recognizes or recalls "ate a hot dog" or "drank a Coke" after attending a baseball game at which a hamburger was eaten and a 7-Up was drunk. Although memory-test instructions warn children against reporting things that were not experienced, false-but-gist-consistent events are often erroneously recognized (Brainerd, Reyna, & Kneer, 1995; Seamon, Luo, Schlegel, Greene, & Goldenberg, 2000), and they sometimes intrude during cued or free recall (e.g., Bjorklund, Bjorklund, Brown, & Cassel, 1998; Bjorklund & Muir, 1988; Brown, Smiley, Day, Townsend, & Lawton, 1977; Howe, 1991). Results of prior research indicate that such responses will occur when children are processing the gist of their experience (e.g., "went to a baseball game") rather than retrieving verbatim traces of actual events (Reyna & Kiernan, 1994, 1995). Of course, false-but-gist-consistent events will seem very familiar when the gist of experience is being processed, and such feelings of familiarity obviously favor the decision that false events were part of the index experience (Brainerd & Reyna, 1998b; Brainerd et al., 1999; Brainerd, Reyna, Wright, & Mojardin, 2001).

However, it may be possible for children to overcome those feelings of familiarity and edit false-but-gist-consistent events out of their memory reports. When such events are presented on recognition tests, or when they come to mind during recall, the usual laws of retrieval, such as encoding variability (e.g., Ackerman, 1985, 1987), predict that these false-but-gist-consistent events will sometimes provoke retrieval of verbatim traces of the corresponding true events (e.g., "hot dog" will sometimes provoke retrieval of verbatim traces of "hamburger," and "Coke" will sometimes provoke retrieval of verbatim traces of "7-Up"). The simultaneous processing of two related items, one of which is known to have been experienced (because a verbatim trace is available) can generate mismatches at the level of verbatim detail. Hot dogs and hamburgers are experienced as similar, and so are Coke and 7-Up, but they also mismatch in palpable ways: Hot dogs and hamburgers look, taste, and smell different, and so do Coke and 7-Up. Verbatim mismatches provide a reliable, positive basis for rejecting false events, a basis that supplies a rationale for how events that were not experienced could nevertheless seem very familiar. ("I ate something at the game, but it wasn't a hot dog because I distinctly remember eating a hamburger.")¹

In studies of spontaneous false memory, recollection rejection, if it occurs, would take place on memory tests when false-but-gist-consistent events are presented as recognition probes or when they come to mind during free or cued recall. Likewise, in studies of memory suggestion, recollection rejection can occur on memory tests, but it can also occur earlier, when misinformation is pre-

¹ Even if the child did not eat a hamburger and drink a 7-Up, there may be other relevant verbatim traces whose retrieval would override the familiarity of "ate a hot dog" and "drank a Coke." For instance, the child might retrieve a verbatim trace of his or her parents saying they would wait and eat at home after the game.

sented to children. According to fuzzy-trace theory, the probability of recollecting a verbatim trace is governed by (a) its accessibility (which is a function of factors such as delay, materials, age, instructions, and prior testing) and (b) the effectiveness of the cue (e.g., memory probe) in evoking memory for original events (Reyna, 1995; Reyna & Titcomb, 1997). Given this mechanism, it is clear that good cues will not yield effective recollection rejection unless verbatim traces are, at least minimally, accessible. When verbatim traces are highly accessible, it has been found that distractors that are related to but that do not match original events are sufficient to evoke them (Brainerd, Reyna, & Kneer, 1995; Brainerd et al., 1999). Note that this mechanism is not a metacognitive strategy based on distinctiveness (although such strategies may operate *in addition to* the basic mechanisms that we discuss here).²

A promising feature of recollection rejection is its developmental plausibility. On the basis of the extant memory development literature (for a review, see Schneider & Bjorklund, 1998), recollection rejection does not involve storage or retrieval skills that are beyond the capabilities of young children. On the contrary, there is independent evidence that those skills are present in young children. On the storage side, young children appear to retain independent representations of the surface form and meaning of word lists (Brainerd, Reyna, & Kneer, 1995), narratives (Reyna & Kiernan, 1994), metaphors (Reyna & Kiernan, 1995), and everyday experiences (Powell, Roberts, Ceci, & Hembroke, 1999). On the retrieval side, if young children are capable of recollection rejection, it follows that false recognition or false recall of gist-consistent events should be reduced by manipulations that increase the accessibility of verbatim traces of actual experience. As noted, one such manipulation, varying the interval between original experience and misinformation, has had the predicted effect in memory suggestion studies (Reyna, 1995). It has also had the predicted effect in studies of spontaneous false memory (Brainerd & Reyna, 1996; Pipe et al., 1999; Poole & White, 1993; Reyna & Kiernan, 1994, 1995). Further, retrieval-cuing manipulations that selectively enhance children's ability to access verbatim traces on memory tests should reduce false reporting, and this prediction, too, has been confirmed (Brainerd & Mojardin, 1998; Brainerd, Reyna, & Kneer, 1995).

To summarize, the process theory of recollection rejection involves the following three components: (a) Verbatim traces of true events are accessible in children's memories; (b) false-but-gist-consistent events sometimes cue the retrieval of verbatim traces of the corresponding true events; and (c) the perception that true events differ in verbatim details from false-but-gist-consistent events overrides the familiarity of the latter, thereby suppressing false reports. We do not assume that recollection rejection requires that any of these components be under deliberate strategic control, although they may be, especially in adults, and levels of recollection rejection may increase when they are. Also, we do not assume that recollection rejection requires participants to be consciously aware of verbatim mismatches between true and false events, although participants can be, especially when they are adults, and such awareness may enhance recollection rejection.

Release From Recollection Rejection: An Experimental Paradigm

Detecting the presence of recollection rejection requires a methodology that implements the theoretical definition of this editing

operation and separates its influences from those of other memory processes. Various methodologies might be possible, but one that ought to be highly sensitive to children's use of recollection rejection combines a memory-testing procedure that was designed to foster recollection rejection (Reyna & Kiernan, 1994, 1995) with the conjoint-recognition model (Brainerd et al., 1999). In Reyna and Kiernan's procedure, children of various ages are exposed to to-be-remembered material that consists of either short literal narratives about everyday events (e.g., "The coffee is hotter than the tea. The tea is hotter than the cocoa. The cocoa is very sweet.") or short metaphorical narratives about everyday events (e.g., "The woman was shopping in the grocery store. The woman saw the lost boy near the door. The woman was an aspirin, kneeling by the lost boy.") The statements in these narratives are individually meaningful, and they also instantiate gists that connect statements (e.g., the first narrative establishes an overall ordering of the three drinks with respect to temperature).

Following presentation of narrative statements, children respond to recognition tests that contain three types of probes: targets, true distractors, and false distractors. Full sets of targets and distractors appear in Table 1 for the example involving temperature relations among the three drinks and for a second example involving spatial relations among some flowers, a table, and a light. As can be seen, targets are statements that have actually been presented. True distractors are *gist-consistent* unpresented statements that either preserve the gist of individual targets (e.g., "The tea is cooler than the coffee.") or that preserve gist that connects targets (e.g., "The coffee is hotter than the cocoa."). False distractors, on the other hand, are *gist-violating* unpresented statements that either violate the gist of individual targets (e.g., "The tea is hotter than the coffee.") or violate gist that connects targets (e.g., "The cocoa is hotter than the coffee.").

The important attribute of Reyna and Kiernan's (1994, 1995) procedure is their testing methodology. Children responded under two types of instructional conditions, which we shall call *verbatim acceptance* and *gist acceptance*. Verbatim acceptance may be thought of as a standard recognition test on which the task is to accept only events that actually happened (targets) and to reject events that did not happen (distractors). Reyna and Kiernan's participants in this condition received instructions and examples to ensure that they understood the distinction between the three types of probes. The instructions included practice narratives, followed by a series of practice probes on which children had to identify whether each probe (a) had been presented, (b) had not been

² Something akin to recollection rejection, which might be confused with it, has long figured in the designs of child and adult misinformation studies. Such studies normally minimize the distinctiveness of the misinformation that is presented on the grounds that misinformation that produces jarring contrasts with experienced events will not increase false reports and may also suppress true reports (e.g., Belli, 1989; Cassel & Bjorklund, 1995). However, this phenomenon differs from recollection rejection in two key respects. First, it involves global suppression of both true and false reports, whereas recollection rejection is narrowly focused on a specific type of false event (namely, false-but-gist-consistent events). Second, because both true and false reports are suppressed, this phenomenon presumably involves application of a metacognitive strategy that centers on informational distinctiveness (e.g., see Dodson & Schacter, in press). As we noted, however, metacognitive strategies are not assumed to be necessary for recollection rejection.

Table 1
Examples of Targets, True Distractors, and False Distractors Used by Reyna and Kiernan (1994)

Item type	Item	
	Example 1	Example 2
Targets		
TPO	The coffee is hotter than the tea.	The flowers are on the table.
TPO	The tea is hotter than the cocoa.	The table is under the light.
TPO	The cocoa is very sweet.	The flowers are in a green pot.
True distractors		
TPN	The tea is cooler than the coffee.	The light is above the table.
TIO	The coffee is hotter than the cocoa.	The flowers are under the light.
TIN	The cocoa is cooler than the coffee.	The light is above the flowers.
False distractors		
FPN	The coffee is cooler than the tea.	The table is above the light.
FIO	The cocoa is hotter than the coffee.	The light is under the flowers.
FIN	The coffee is cooler than the cocoa.	The flowers are above the light.

Note. TPO = true premise with old words, TPN = true premise with novel words, TIO = true inference with old words, TIN = true inference with novel words, FPN = false premise with novel words, FIO = false inference with old words, and FIN = false inference with new words.

presented and was true, or (c) had not been presented and was false. This procedure was followed to ensure that children understood that true distractors should be rejected when they failed to match the exact wording of presented sentences (recollection rejection), whereas false distractors should be rejected because they failed to match exact wording or they violated narrative gist. Children were required to successfully complete the instructional phase in order to participate in the experiment. Children as young as 5 and 6 years (the youngest age level tested) were able to do so without difficulty.

The gist-acceptance condition, in contrast, may be thought of as a meaning-recognition test on which the task is to accept events that could have happened (targets and true distractors), irrespective of whether they were explicitly presented, and to reject events that could not have happened (false distractors). The children in this condition also received detailed instructions and completed a practice phase to ensure that they understood the distinction between the two types of probes. The instructions included the same practice narratives followed by the same series of probes as in the verbatim-acceptance condition. However, the children's task was simply to identify which probes were true and which probes were false. On the basis of the instructions, children understood that all probes that preserved narrative gist were to be accepted, regardless of whether they matched the exact wording of presented sentences, and that all probes that violated narrative gist were to be rejected. The crucial feature of the gist-acceptance condition is that it releases children from the need to rely on the recollection rejection operation because surface mismatch does not necessarily mean that a probe is to be rejected.

Thus, there is a single, key difference between the verbatim- and gist-acceptance conditions: Verbatim mismatch supports rejection decisions in the former condition, but it does not support any particular decision in the latter condition. This difference can be exploited to measure recollection rejection by using the conjoint-recognition model (Brainerd et al., 1999; Brainerd, Stein, & Reyna, 1998; Reyna & Brainerd, 1998) to express the acceptance probabilities for gist-consistent and gist-violating distractors as functions of verbatim retrieval, gist retrieval, and response bias. In

the conjoint-recognition model, distractors such as those in Table 1 are assumed (a) to provoke retrieval of corresponding verbatim traces and to generate verbatim mismatches with probability V and (b) to provoke retrieval of corresponding gist traces and to generate meaning matches or mismatches with probability G . The parameter V is of central interest, of course, because it indexes the process of recollection rejection. When a distractor fails to produce either form of retrieval, recognition responses are based on guessing or some other form of response bias, yielding acceptance with some probability β and rejection with probability $1 - \beta$. Thus, the probability of falsely accepting (A) a gist-consistent distractor in the verbatim-acceptance condition is

$$p_{VA,T}(A) = (1 - V)G + (1 - V)(1 - G)\beta_{VA}, \quad (1)$$

and the probability of correctly rejecting (R) it is

$$p_{VA,T}(R) = V + (1 - V)(1 - G)(1 - \beta_{VA}), \quad (2)$$

where the subscript T denotes a "true" (gist-consistent) distractor and the subscript VA denotes the verbatim-acceptance condition. Similarly, the probability of falsely accepting a gist-violating distractor is

$$p_{VA,F}(A) = (1 - V)(1 - G)\beta_{VA}, \quad (3)$$

and the probability of correctly rejecting it is

$$p_{VA,F}(R) = V + (1 - V)G + (1 - V)(1 - G)(1 - \beta_{VA}), \quad (4)$$

where the subscript F denotes a "false" (gist-violating) distractor. Equations 1 and 2 merely formalize the notion that a gist-consistent distractor will produce false alarms if verbatim processing fails and gist processing succeeds or if response bias generates acceptance when verbatim and gist processing both fail. Equations 3 and 4 formalize the notion that gist-violating distractors will produce false alarms only when verbatim and gist processing both fail and response bias generates acceptance.

With regard to the gist-acceptance condition, the probability of correctly accepting a gist-consistent distractor is

$$p_{GA,T}(A) = V + (1 - V)G + (1 - V)(1 - G)\beta_{GA}, \quad (5)$$

and the probability of incorrectly rejecting it is

$$p_{GA,T}(R) = (1 - V)(1 - G)(1 - \beta_{GA}), \quad (6)$$

where the subscript T again denotes a “true” (gist-consistent) distractor and the subscript GA denotes the gist-acceptance condition. Similarly, the probability of falsely accepting a gist-violating distractor is

$$p_{GA,F}(A) = (1 - V)(1 - G)\beta_{GA}, \quad (7)$$

and the probability of correctly rejecting it is

$$p_{GA,F}(R) = V + (1 - V)G + (1 - V)(1 - G)(1 - \beta_{GA}), \quad (8)$$

where the subscript F again denotes a “false” (gist-violating) distractor. Note that the response-bias parameter under gist-acceptance instructions (β_{GA}) is not assumed to be the same as the response-bias parameter under verbatim-acceptance instructions (β_{VA}). Prior research shows that bias is usually higher in conditions that allow more types of items to be accepted (i.e., targets + gist-consistent distractors) than in conditions that allow fewer types of items to be accepted (i.e., targets only).

In these expressions, V , G , β_{VA} , and β_{GA} all refer to memory processes—specifically, to verbatim processing, gist processing, and response bias—not to children’s actual performance on recognition tests. However, all of these memory processes can be estimated from data by merely solving the above equations so that each memory process is expressed in terms of acceptance probabilities, rather than vice versa. First, Equation 1 can be rewritten as

$$p_{VA,T}(A) = (1 - V)G + p_{VA,F}(A), \quad (9)$$

which eliminates the bias parameter β_{VA} , and Equation 5 can be rewritten as

$$p_{GA,T}(A) = V + (1 - V)G + p_{GA,F}(A), \quad (10)$$

which eliminates the bias parameter β_{GA} . Second, subtraction of Equation 9 from Equation 10 yields

$$p_{GA,T}(A) - p_{VA,T}(A) = V + p_{GA,F}(A) - p_{VA,F}(A), \quad (11)$$

which can be rearranged to produce

$$V = [p_{GA,T}(A) - p_{VA,T}(A)] + [p_{VA,F}(A) - p_{GA,F}(A)]. \quad (12)$$

Thus, the process of central interest, recollection rejection, can be directly estimated from the performance data that are generated by Reyna and Kiernan’s (1994, 1995) procedure by combining the acceptance probabilities for gist-consistent and gist-violating distractors across the two conditions.

Although principal interest attaches to measuring recollection rejection, the gist-memory process and the two bias parameters can also be estimated by working backward from Equation 12. Once V has been estimated for a data sample, G can then be estimated from

$$G = [p_{VA,T}(A) - p_{VA,F}(A)] \div V. \quad (13)$$

Finally, the two bias parameters can be estimated from

$$\beta_{VA} = p_{VA,F}(A) \div [(1 - V)(1 - G)] \quad (14)$$

and

$$\beta_{GA} = p_{GA,F}(A) \div [(1 - V)(1 - G)]. \quad (15)$$

It might be thought that the model in Equation 12 is unnecessary and that, instead, recollection rejection can be studied by simply comparing children’s acceptance of gist-consistent distractors under verbatim instructions versus gist instructions. Because retrieval of verbatim traces supports rejection under verbatim instructions but not gist instructions, acceptance rates for gist-consistent distractors should be higher in the latter condition, and it may therefore seem natural to assume that the difference between the two conditions is a measure of recollection rejection. This assumption is an example of a common misconception in psychology, the process-purity fallacy (e.g., Howe, Rabinowitz, & Grant, 1993; Yu & Bellezza, 2000). The assumption that a memory process can be measured by merely comparing performance in two different testing conditions is valid only if the difference between them is entirely due to a single process (recollection rejection, in this instance), which is rarely the case. As far as recollection rejection, in particular, is concerned, if we subtract Equation 1 from Equation 4, the result is

$$p_{GA,T}(A) - p_{VA,T}(A) = V + (1 - V)(1 - G)[\beta_{GA} - \beta_{VA}], \quad (16)$$

from which it can be seen that the difference between these conditions depends on response bias and gist retrieval, as well as on verbatim retrieval.

To summarize the material in the present section, we have described a paradigm that can be used to detect the use of recollection rejection to suppress reports of false-but-gist-consistent information. That paradigm, which we call release from recollection rejection, has two basic features: Memory tests are administered under different instructions (verbatim acceptance and gist acceptance), and a simple mathematical model is defined over these conditions that estimates recollection rejection (and other relevant memory processes).³ In the next section, we exploit this paradigm to provide preliminary data on children’s use of recollection rejection.

Measurement of Recollection Rejection in Children

We now apply the above techniques to three previously published data sets. It is possible to do this because, in each data set, children responded to recognition tests under both verbatim-acceptance and gist-acceptance instructions. Although our first concern is with the measurement of recollection rejection, the interpretation of the resulting data is strengthened by collateral

³ As with all mathematical models, the mapping of posited memory processes (recollection rejection, gist-based acceptance, and response bias in this instance) onto the present model’s parameters (V , G , and β) is not necessarily one to one (Brainerd, Reyna, Wright, & Payne, in press). More explicitly, although the model that we use to measure recollection rejection is implied by fuzzy-trace theory’s definition of this editing operation, other theoretical conceptions might imply the same model. When different process interpretations of a model’s parameters exist, or when a new interpretation of an existing model is proposed, there is an established method of adjudication—namely, to determine through experimentation how the parameters react to manipulations that the different interpretations make contrasting predictions about (Brainerd, Howe, & Desrochers, 1982).

evidence validating the recollection-rejection model. With models such as this, two types of validity evidence are relevant (see Brainerd, Howe, & Desrochers, 1982): (a) evidence showing that the model provides acceptable fits to the data from which the measurements are derived and (b) evidence showing that the model's parameters react sensibly to experimental manipulations. Consequently, in addition to measuring recollection rejection, our analyses focus on goodness of fit and on whether the model's parameters—especially V , the parameter that indexes recollection rejection—respond appropriately to manipulations that should affect the difficulty of the processes that they are presumed to measure.

Reyna and Kiernan (1994)

Reyna and Kiernan (1994) reported two experiments in which 50 younger children (6-year-olds) and 50 older children (9-year-olds) listened to eight literal narratives like those described earlier, four of which specified linear relations among objects (see Table 1, Example 1) and four of which specified spatial relations (see Table 1, Example 2). Recognition tests were administered for targets, gist-consistent distractors, and gist-violating distractors like those in Table 1. There were two tests, the first immediately after narrative presentation and the second 1 week later. Half the children at each age level responded under verbatim-acceptance instructions, and half responded under gist-acceptance instructions.

Overall results for recollection rejection. To provide an overall picture of children's use of recollection rejection, we used Equations 12–16 to estimate the parameters V , G , β_{VA} , and β_{GA} for Reyna and Kiernan's (1994) participants. This was done as follows. Note in Table 1 that three different types of gist-consistent distractors were administered (true premise with novel words, true inference with old words, and true inference with novel words, denoted as TPN, TIO, and TIN, respectively), each of which is paired with a gist-violating distractor that contains the same words but has a different meaning (false premise with novel words, false inference with old words, and false inference with novel words, denoted as FPN, FIO, and FIN, respectively). For instance, "The tea is cooler than the coffee" (TPN) is paired with "The coffee is cooler than the tea" (FPN); "The coffee is hotter than the cocoa" (TIO) is paired with "The cocoa is hotter than the coffee" (FIO); and "The cocoa is cooler than the coffee" (TIN) is paired with "The coffee is cooler than the cocoa" (FIN). The parameters V , G , β_{VA} , and β_{GA} were estimated separately for each of these pairings of gist-consistent and gist-violating distractors. Further, the parameters for each pairing were estimated separately for each age level (younger vs. older) and each memory test (immediate vs. delayed). Thus, 12 sets of parameter estimates were generated for these data (3 distractor pairings $\times$ 2 age levels $\times$ 2 tests). The overall findings were as follows.

For younger children, pooling across the three distractor pairings on the immediate test, the mean values of the four parameters were $V = .15$, $G = .24$, $\beta_{VA} = .37$, and $\beta_{GA} = .49$. The estimate of V shows that younger children were able to use verbatim mismatch roughly 15% of the time to suppress acceptance of gist-consistent distractors. The estimate of G indicates that these distractors were more apt to provoke gist processing than verbatim processing (24% vs. 15%). For older children, pooling across the three distractor pairings on the immediate test, the mean values of

the four parameters were $V = .39$, $G = .31$, $\beta_{VA} = .22$, and $\beta_{GA} = .52$. The estimate of V shows that older children used verbatim mismatch to suppress acceptance of gist-consistent distractors roughly 39% of the time, more than twice as often as younger children. This age increase in verbatim processing (parameter V) was significant, $G^2(1) = 35.71$, $p < .01$. (Statistical procedures for testing between-condition hypotheses about observed values of the model's parameters are described in the Appendix, in connection with Equation A13.) The estimate of G shows that unlike the case with younger children, these distractors were more apt to provoke verbatim processing than gist processing (39% vs. 31%). The age increase in gist processing (parameter G) was also significant, $G^2(1) = 49.36$, $p < .01$.

It is instructive to compute the proportion of correct rejections of gist-consistent distractors (in the verbatim-acceptance condition) that were due to verbatim mismatch versus other reasons. (Technically, this proportion is the conditional probability that a distractor is rejected through verbatim mismatch given that it is rejected at all.) According to Equation 2, such distractors are correctly rejected for two reasons—namely, verbatim mismatch (with probability V) and "no memory" (with probability $[1 - V][1 - G][1 - \beta_{VA}]$). Although the former qualifies as false-memory editing (specific distractors are excluded owing to surface mismatches with specific verbatim traces), the latter is simply passive, generic failure to accept because of inadequate memory support. On the immediate test, the pooled correct rejection rates for gist-consistent distractors were .57 and .71 for younger and older children, respectively, so the corresponding proportions of rejections from verbatim mismatch were $.15 \div .57 = .26$ and $.39 \div .71 = .55$. Hence, in addition to the developmental increase in absolute levels of recollection rejection, there was a doubling of the chance that a correct rejection of a gist-consistent distractor was due to verbatim mismatch, and by age 9, verbatim mismatches accounted for more than half of such rejections.

The mean false-alarm rate for gist-consistent distractors in the verbatim-acceptance condition declined with age, from .43 in 6-year-olds to .29 in 9-year-olds. On the basis of the preceding parameter estimates, there were two underlying causes of this developmental decline: Children's ability to generate verbatim mismatches more than doubled with age, from .15 to .39, and their levels of response bias decreased by nearly half, from .37 to .22. [The decrease in response bias was reliable, $G^2(1) = 61.24$, $p < .01$.] It is important to recognize that the developmental decline in false memory was not caused by decreases in gist processing (which supports false alarms). As previously mentioned, children's processing of narrative gist on recognition tests, as measured by the parameter G , also increased with age (from .24 to .31). Therefore, the decline in false alarms in response to gist-consistent distractors occurred despite the fact that support for such errors from gist processing increased with age (for similar arguments, see Reyna & Kiernan, 1994). This pattern of developmental findings for verbatim processing, gist processing, and response bias underscores a key advantage of using a multiprocess model to decompose age trends in raw memory performance: Age trends in raw performance can occur even though changes in some underlying processes militate against them, and age trends in raw performance can fail to occur because changes in underlying processes that have opposite effects on raw performance cancel each other out (Brainerd, 1985).

On the delayed test, age differences disappeared. It is important to note that the false-alarm rate for gist-consistent distractors (in the verbatim-acceptance condition) remained stable across the 1-week interval in younger children but increased (from .29 to .45) in older children. Pooling across the three types of distractor pairings and the two age levels, the mean values of the four parameters on the delayed test were $V = .06$, $G = .17$, $\beta_{VA} = .38$, and $\beta_{GA} = .50$. One week after narrative presentation, therefore, the mean level of recollection rejection had decreased by 78% (from .27 to .06), whereas the mean level of gist processing had decreased by 44% (from .30 to .17). Further, the mean proportion of correct rejections of gist-consistent distractors that were due to verbatim mismatches decreased by 68% (from .41 to .13). The proportion of correct rejections that were due to verbatim mismatches decreased from an average of 64% to 29% on the delayed test, so that rejection had become largely a matter of passive nonacceptance because of a lack of memory support. Of course, this dramatic decline over a delay in children's ability to use verbatim mismatch to suppress false reports of gist-consistent distractors is consistent with the well-known finding that verbatim memories become rapidly inaccessible with time whereas gist memories are more robust (e.g., Gernsbacher, 1985; Kintsch, Welch, Smalhofer, & Zimny, 1990).

Model validity results. We turn now to the deeper question of the validity of the recollection-rejection model as an experimental tool for studying children's false-memory editing. As noted earlier, two types of findings bear on this question: goodness of fit and parameter behavior. Concerning fit, models that cannot predict actual data—in this case, release-from-recollection-rejection data—can be dropped without further consideration. Concerning parameter behavior, a model's parameters should react in ways that are consistent with their theoretical definitions. That is, parameters should react sensibly to experimental manipulations that affect the processes that they ostensibly measure. With the present model, V should react appropriately to manipulations that affect the difficulty of verbatim processing, and G should react appropriately to manipulations that affect the difficulty of gist processing.

If we take goodness of fit first, two types of fit tests are possible: qualitative and quantitative. The qualitative test is concerned with the difference between acceptance rates for gist-consistent and gist-violating distractors in the verbatim-acceptance condition versus the gist-acceptance condition. For the verbatim-acceptance condition, this difference is found by subtracting Equation 3 from Equation 1—specifically, $[(1 - V)G + (1 - V)(1 - G)\beta_{VA}] - [(1 - V)(1 - G)\beta_{VA}]$. For the gist-acceptance condition, the same difference is found by subtracting Equation 7 from Equation 5—specifically, $[V + (1 - V)G + (1 - V)(1 - G)\beta_{GA}] - [(1 - V)(1 - G)(1 - \beta_{GA})]$. Obviously, the latter difference must be larger than the former as long as recollection rejection occurs (i.e., as long as $V > 0$). Therefore, the recollection-rejection model makes the straightforward prediction that the difference between the acceptance rates for gist-consistent and gist-violating distractors must be greater in the gist-acceptance condition than in the verbatim-acceptance condition except in the limiting condition where recollection rejection does not occur. The relevant data for Reyna and Kiernan's (1994) immediate test (where verbatim traces should still be accessible) are displayed in Figure 1 for both younger (Panel A) and older (Panel B) children. It can be seen that the prediction was confirmed at both age levels.

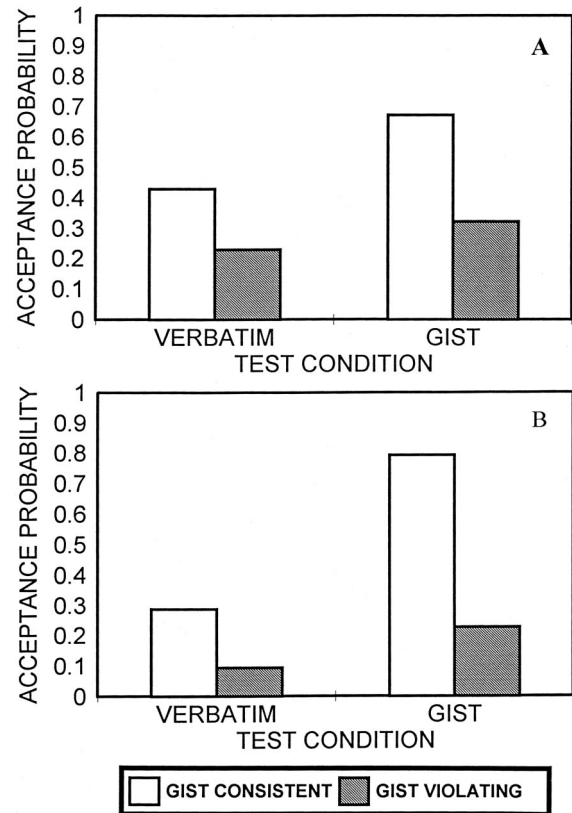


Figure 1. For Reyna and Kiernan's (1994) experiments, mean acceptance rates for gist-consistent and gist-violating distractors in the verbatim-acceptance and gist-acceptance conditions for 6-year-olds (Panel A) and 9-year-olds (Panel B).

With regard to quantitative fit, the appropriate fit test for the recollection-rejection model is given in the Appendix (Equation A11). This test was conducted for 12 data sets (3 distractor pairings $\times$ 2 age levels $\times$ 2 memory tests). For each data set, we conducted the test by implementing the recollection-rejection model (specifically, Equations A1–A8) in Hu's (1995) General Processing Tree (GPT) software and then computing the fit statistic in Equation A11. Across the 12 data sets, the average computed value of the statistic was 2.42, which is less than half the value that is required to reject the null hypothesis of fit at the .05 level (5.99). Thus, the model gave an excellent account of Reyna and Kiernan's (1994) data.

For the second validity criterion, parameter behavior, two results of this sort have already been reported: Consistent with the notion that verbatim memories fade more rapidly than gist memories, the proportionate decline in V over the forgetting interval was nearly twice the proportionate decline in G , and the proportion of correct rejections of gist-consistent distractors (in the verbatim-acceptance condition) that were due to verbatim mismatches declined by more than 50%. Further results that bear on the second model-validity criterion come from (a) developmental trends in the parameters, (b) a comparison of levels of recollection rejection for the three types of gist-consistent distractors (TPN, TIO, and TIN), and (c) a comparison of response-bias levels for the verbatim-acceptance and gist-acceptance conditions.

Concerning developmental trends, other research, such as Brainerd, Reyna, and Kneer's (1995) data on the phenomenon of false-recognition reversal, suggests that the ability to access verbatim traces of experience increases during childhood (Reyna, 1996). Therefore, the value of V should increase with age for each of the three types of gist-consistent distractors. Consistent with this prediction, the value of V was larger for 9-year-olds with TPN distractors, $G^2(1) = 25.05$, $p < .01$, with TIO distractors, $G^2(1) = 26.31$, $p < .01$, and with TIN distractors, $G^2(1) = 27.32$, $p < .0001$. Prior research (e.g., Brainerd & Mojardin, 1998) also suggests that on standard memory tests, in which children are told to accept only material that was actually presented, response bias levels usually decrease with age. Consistent with this prediction, the value of the response-bias parameter in the verbatim-acceptance condition (β_{VA}) was smaller for 9-year-olds than for 6-year-olds with TPN distractors, $G^2(1) = 26.37$, $p < .0001$, with TIO distractors, $G^2(1) = 29.02$, $p < .01$, and with TIN distractors, $G^2(1) = 38.29$, $p < .01$.

Concerning levels of recollection rejection for the three types of gist-consistent distractors, according to the theoretical definition of recollection rejection, children are able to reject such distractors because they retrieve verbatim traces, which generates mismatches between those traces and the distractors' surface forms. The crucial step that sets this process in motion, verbatim retrieval, ought to be easier for TIO and TIN distractors than for TPN distractors. TIO and TIN distractors are both constructed by combining the first half of one target with the second half of another target (see Table 1 for examples). Therefore, retrieval of the verbatim trace of either of these targets (or both of them) will generate a verbatim mismatch. In contrast, a TPN distractor is a paraphrase of a single target, and therefore only retrieval of the verbatim trace of that target will generate a verbatim mismatch. (In making this prediction, it is assumed that although factors other than verbatim retrieval contribute to recollection rejection, those factors are the same for TPN, TIO, and TIN distractors.) Consistent with this prediction, on the immediate test, the value of V for 6-year-olds was smaller for TPN than for TIO distractors, $G^2(1) = 21.27$, $p < .01$, and it was smaller for TPN than for TIN distractors, $G^2(1) = 27.32$, $p < .01$. Similarly, on the immediate test, the value of V for 9-year-olds was smaller for TPN than for TIO distractors, $G^2(1) = 22.16$, $p < .01$, and it was smaller for TPN than for TIN distractors, $G^2(1) = 19.02$, $p < .01$. Although levels of recollection rejection differed in the predicted manner for the three types of gist-consistent distractors, parallel variation in levels of gist processing (parameter G) would not be expected because all of these distractors are faithful to narrative gist. Consistent with this prediction, the value of the parameter G did not differ significantly among the three types of gist-consistent distractors, with the mean estimates being .27 (TPN), .32 (TIO), and .25 (TIN).

Last, concerning response-bias levels for verbatim acceptance versus gist acceptance, two predictions can be made on the basis of prior research that provide further validity tests. First, as mentioned above, a consistent finding about response bias is that it tends to be higher in conditions that allow more types of test items to be accepted (e.g., Brainerd et al., 1999; Buchner, Erdfelder, Vatervodt-Pluncke, 1995). In Reyna and Kiernan's (1994) experiments, therefore, response bias levels should be higher in the gist-acceptance condition (parameter β_{GA}) than in the verbatim-acceptance condition (parameter β_{VA}). For 6-year-olds, this prediction was confirmed with TPN distractors, $G^2(1) = 20.69$, and

TIN distractors, $G^2(1) = 9.02$, $p < .01$, but not with TIO distractors (though the difference was in the predicted direction). For 9-year-olds, the prediction was confirmed with TPN distractors, $G^2(1) = 11.18$, $p < .01$, with TIO distractors, $G^2(1) = 15.01$, $p < .01$, and with TIN distractors, $G^2(1) = 11.40$, $p < .01$. Overall, in agreement with prediction, the average value of β_{GA} (.51) was nearly twice the average value of β_{VA} (.29).

The second prediction about response bias is another familiar expectation about such parameters—namely, that on standard memory tests, in which children are told to accept only material that was actually presented, response bias tends to increase as time passes (see Brainerd, Reyna, & Kneer, 1995). Consistent with this prediction, response bias in the verbatim-acceptance condition (parameter β_{VA}) increased between the immediate and 1-week tests. The increase was significant for both 6-year-olds, $G^2(1) = 147.59$, $p < .01$, and 9-year-olds, $G^2(1) = 62.92$, $p < .01$.

Kiernan (1993)

Two experiments by Kiernan (1993) that used the same materials with different samples of children provide replications of most of the preceding findings. Kiernan's procedures were the same as Reyna and Kiernan's (1994) except for three features. First, the developmental manipulation involved developmental delay rather than chronological age. Kiernan administered Reyna and Kiernan's narratives (with minor modifications) to 16 children (mean age = 9 years 8 months) with clinical diagnoses of delayed language development, which included a score at least 1.5 standard deviations below the mean for their age level on the Clinical Evaluation of Language Fundamentals—Revised (Semel, Wiig, & Secord, 1987), and to 16 matched control children (mean age = 9 years 8 months) whose language development was normal. Levels of recollection rejection would be expected to be lower in the language-delayed children than in the control children: Children with clinical language delay exhibit deficits in memory for the exact surface form of verbal material (e.g., Menyuk, 1964). Second, the number of narratives that was administered to each child was increased from 8 to 16 (8 spatial and 8 linear). Third, only two of Reyna and Kiernan's gist-consistent distractors (TPN and TIO) were administered.

Overall results for recollection rejection. We used Equations 12–16 to estimate V , G , β_{VA} , and β_{GA} for Kiernan's (1993) participants following the same procedure that was described earlier. Unlike Reyna and Kiernan (1994), Kiernan (1993) did not report data in such a way that the quantitative goodness-of-fit test in Equation A11 could be computed. This is not a serious limitation because that test was successful for these same procedures when it was computed for Reyna and Kiernan's experiments and because the qualitative test that we described earlier can be computed. Recall that the qualitative test evaluates the hypothesis that the difference between the acceptance rates for gist-consistent and gist-violating distractors must be greater in the gist-acceptance condition than in the verbatim-acceptance condition. The relevant data for Kiernan's immediate test (in which verbatim traces should still be accessible) are displayed in Figure 2 for both language-delayed (Panel A) and control (Panel B) children. It can be seen that the prediction was confirmed for both groups of children.

Also unlike Reyna and Kiernan (1994), Kiernan (1993) did not report data in such a way that the parameters could be separately estimated for each of the 16 cells in her factorial design (2

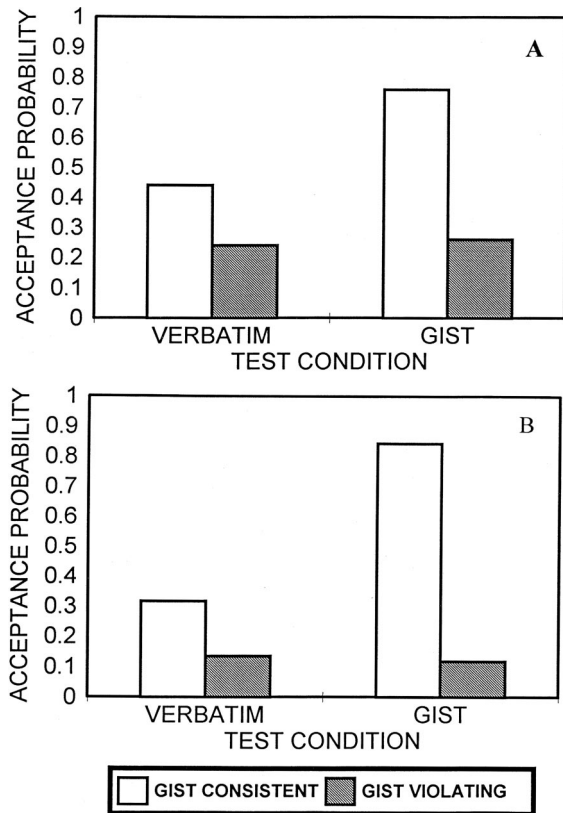


Figure 2. For Kiernan's (1993) experiments, mean acceptance rates for gist-consistent and gist-violating distractors in the verbatim-acceptance and gist-acceptance conditions for language-delayed children (Panel A) and control children (Panel B).

developmental levels [delayed vs. normal] $\times$ 2 types of gist-consistent distractors [TPN vs. TIO] $\times$ 2 types of narrative content [spatial vs. linear] $\times$ 2 testing sessions [immediate vs. delayed]. Rather, she reported performance for verbatim-acceptance and gist-acceptance conditions (a) separately by developmental level and distractor type, pooled across narrative content, for the immediate testing session only and (b) separately by distractor type, narrative content, and testing session, but pooled across developmental level. We present findings for recollection rejection separately for these two methods of data reporting. We do not present further model-validity findings for the response-bias patterns because all the patterns reported earlier were replicated in Kiernan's data.

Developmental Level $\times$ Distractor Type for the immediate testing session (pooled across narrative content). On the immediate test, the mean values of the four parameters for control children were $V = .54$, $G = .39$, $\beta_{VA} = .44$, and $\beta_{GA} = .44$. The corresponding values for language-delayed children were $V = .26$, $G = .27$, $\beta_{VA} = .44$, and $\beta_{GA} = .48$. Hence, control children were able to use verbatim mismatch to suppress acceptance of gist-consistent distractors roughly twice as often as were language-delayed children, with the difference being significant for both TPN distractors, $G^2(1) = 35.01$, $p < .01$, and TIO distractors, $G^2(1) = 35.20$, $p < .01$. There was also a dramatic developmental difference in the proportion of correct rejections that were due to

verbatim mismatches. Pooled across distractors, the average probability of a correct rejection in the verbatim-acceptance condition by control children was .68, so that verbatim mismatches accounted for 79% of those responses. For language-delayed children, the average probability of a correct rejection was .56, so that verbatim mismatches accounted for only about half as many (46%) of those responses. Thus, there was a developmental difference in the tendency for rejections of gist-consistent distractors to be due to recollection rejection rather than to passive nonacceptance from inadequate memory support.

The gist processing parameter (G) also exhibited developmental differences, but the difference between the mean estimates of G for control and language-delayed children (.12) was less than half the difference between the mean estimates of V (.28). The developmental difference in G was significant for TIO distractors, $G^2(1) = 36.97$, $p < .01$, but not for TPN distractors. As before, this developmental difference occurred despite the fact that support for such errors from gist processing was greater in control children than in language-delayed children. Therefore, if the difference in false-alarm rates is to be correctly interpreted, the difference in G , as well as V , must be considered.

Distractor Type $\times$ Narrative Content $\times$ Testing Session (pooled across developmental level). Mean estimates of the four parameters for distractor types, types of narrative content, and testing sessions are reported in Table 2. The results for immediate versus delayed testing preserved the patterns in Reyna and Kiernan's (1994) data in that there was a greater decline in recollection rejection (from .43 to .24) than in gist processing, which actually increased (from .36 to .56). (Increases in gist processing over delays of a few days to a week have been detected in some prior experiments [e.g., Lampinen, Copeland, & Neuschatz, 2001].) Consistent with Reyna and Kiernan's data, then, changes in the parameters V and G in Kiernan's (1993) data concur with the familiar finding that verbatim memories become inaccessible more rapidly than do gist memories.

Table 2
Estimates of the Parameters of the Recollection-Rejection Model for Some of the Conditions of Kiernan's (1993) Experiments

Distractor type/narrative content	Testing session	
	Immediate	Delayed
TPN/spatial		
V	.41	.30
G	.31	.41
β_{VA}	.37	.39
β_{GA}	.35	.39
TPN/linear		
V	.42	0
G	.26	.59
β_{VA}	.35	.50
β_{GA}	.42	.88
TIO/spatial		
V	.37	.39
G	.48	.36
β_{VA}	.21	.41
β_{GA}	.43	.54
TIO/linear		
V	.50	.27
G	.40	.52
β_{VA}	.24	.54
β_{GA}	.50	.54

There was one result for recollection rejection that did not replicate across the Reyna and Kiernan (1994) and Kiernan (1993) data sets—namely, differences for TPN distractors versus TIO distractors. As noted earlier, target–distractor surface dissimilarity is greater for TIO than for TPN distractors, which may make verbatim mismatches more perceptible in the former case. Consistent with this notion, the estimate of V was indeed larger for TPN distractors than for TIO distractors in Reyna and Kiernan’s experiments. However, this difference was not significant in Kiernan’s (1993) experiments. Given the relative comparability of experimental procedures, there is no obvious explanation of this failure to replicate, but because Kiernan’s subject sample was small, it may simply be a statistical power problem.

Reyna and Kiernan (1995)

Reyna and Kiernan (1995) reported two experiments that paralleled those of their earlier study (Reyna & Kiernan, 1994). As before, 50 younger children (6-year-olds) and 50 older children (9-year-olds) listened to a total of eight narratives, each composed of three sentences, which were followed by recognition tests for targets, gist-consistent distractors, and gist-violating distractors. However, the narratives contained metaphorical statements about familiar objects and events. An example is shown in Table 3. The first and second sentences in each narrative were literal (“The woman was shopping in the grocery store. The woman saw the lost boy near the door.”), but the third sentence was metaphorical (“The woman was an aspirin, kneeling by the lost boy.”) Three pairs of gist-consistent and gist-violating distractors were administered on the recognition tests (TLIT–FLIT, TPER–FPER, and TPSY–FPSY). The gist-consistent distractors were as follows: TLIT = statements that were paraphrases of the meaning of metaphorical targets (“The woman was medicine, kneeling by the

lost boy.”), TPER = statements that were true perceptual descriptions of a feature of metaphorical targets (“The woman was white and round, kneeling by the lost boy.”), TPSY = statements that were true psychological interpretations of metaphorical targets (“The woman made him feel better, kneeling by the lost boy.”). The gist-violating distractors were as follows: FLIT = statements that were false paraphrases of the meaning of metaphorical targets (“The woman was a nurse, kneeling by the lost boy.”), FPER = statements that were false perceptual descriptions of a feature of metaphorical targets (“The woman had red cheeks, kneeling by the lost boy.”), and FPSY = statements that were false psychological interpretations of metaphorical targets (“The woman was proud, kneeling by the lost boy.”). As before, recognition tests were administered immediately after narrative presentation and were repeated 1 week later.

Overall results for recollection rejection. We again used Equations 12–16 to estimate the parameters V , G , β_{VA} , and β_{GA} . The parameters were separately estimated for each of the three pairings of gist-consistent and gist-inconsistent distractors (TLIT–FLIT, TPER–FPER, TPSY–FPSY), for each age level (younger vs. older), and for each memory test (immediate vs. delayed), yielding 12 sets of parameter estimates, which are presented in Table 4.

For younger children on the immediate test, the mean values of the four parameters were $V = .08$, $G = .10$, $\beta_{VA} = .09$, and $\beta_{GA} = .40$. Therefore, younger children were able to use verbatim mismatch roughly 8% of the time to suppress acceptance of gist-consistent distractors. Based on the estimate of G , these distractors provoked gist processing in younger children at about the same rate as verbatim processing (10% vs. 8%). For older children on the immediate test, the mean values of the four parameters were $V = .45$, $G = .09$, $\beta_{VA} = .03$, and $\beta_{GA} = .18$. Therefore, older children were able to use verbatim mismatch nearly half of the time (45%) to suppress acceptance of gist-consistent distractors. Based on the estimate of G , these distractors provoked gist processing in older children far less often than verbatim processing (9% vs. 45%). The age increase in verbatim processing (parameter V) was significant for TLIT distractors, $G^2(1) = 40.37$, $p < .01$, TPER distractors, $G^2(1) = 20.71$, $p < .01$, and TPSY distractors, $G^2(1) = 25.06$, $p < .01$.

We again computed the proportion of correct rejections of gist-consistent distractors (in the verbatim-acceptance condition) that were due to verbatim mismatch versus other reasons. On the immediate test, the proportions of rejections that were due to verbatim mismatch were $.08 \div .83 = .10$ and $.45 \div .93 = .48$ for younger and older children, respectively. Thus, there was nearly a fivefold increase in the chance that correct rejection of a gist-consistent distractor was due to verbatim mismatch, and by age 9, verbatim mismatches accounted for almost half of such rejections.

The mean false-alarm rate for gist-consistent distractors in the verbatim-acceptance condition again declined with age, from .17 to .07. As before, there were two causes: a large age increase in children’s ability to generate verbatim mismatches, from .08 to .45, coupled with a small decrease in response bias (parameter β_{VA}), from .09 to .03. Once again, the age decline in false alarms was not caused by decreases in gist processing (which supports false alarms). Children’s processing of the gist of metaphorical statements on recognition tests, as measured by the parameter G , was virtually the same for younger and older children (.10 vs. .09).

Table 3
Examples of Targets, True Distractors, and False Distractors From Reyna and Kiernan (1995)

Item type	Example
Targets	The woman was shopping in the grocery store. The woman saw the lost boy near the door. The woman was an aspirin, kneeling by the lost boy.
True distractors	
TLIT	The woman was medicine, kneeling by the lost boy.
TPER	The woman was white and round, kneeling by the lost boy.
TPSY	The woman made him feel better, kneeling by the lost boy.
False distractors	
FLIT	The woman was a nurse, kneeling by the lost boy.
FPER	The woman had red cheeks, kneeling by the lost boy.
FPSY	The woman was proud, kneeling by the lost boy.

Note. TLIT = a true paraphrase of the metaphorical target (i.e., aspirin = medicine); TPER = a true perceptual interpretation of a feature of the metaphorical target (i.e., aspirin = white and round); TPSY = a true psychological interpretation of the metaphorical target (i.e., the woman would make the boy feel better); FLIT = a false paraphrase of the metaphorical target (i.e., aspirin $\neq$ a nurse); FPER = a false perceptual interpretation of a feature of the metaphorical target (i.e., the woman did not have red cheeks); FPSY = a false psychological interpretation of the metaphorical target (i.e., the woman did not feel proud).

Table 4
Estimates of the Parameters of the Recollection-Rejection Model for the Conditions of Reyna and Kiernan's (1995) Experiments

Distractor	Age level	
	6-year-olds	9-year-olds
Immediate test		
TLIT		
V	.08	.52
G	.13	.16
β_{VA}	.05	.09
β_{GA}	.29	.37
TPER		
V	.01	.39
G	.16	.02
β_{VA}	.09	.02
β_{GA}	.37	.12
TPSY		
V	.15	.44
G	.02	0
β_{VA}	.02	.04
β_{GA}	.12	.04
Delayed test		
TLIT		
V	0	0
G	.23	.25
β_{VA}	.14	.30
β_{GA}	.20	.04
TPER		
V	0	0
G	.07	.03
β_{VA}	.30	.30
β_{GA}	.18	.07
TPSY		
V	.07	0
G	.10	.25
β_{VA}	.50	.55
β_{GA}	.29	.10

Note. TLIT = a true rendering of the meaning of the metaphorical target (i.e., aspirin = medicine); TPER = a true perceptual interpretation of a feature of the metaphorical target (i.e., aspirin = white and round); TPSY = a true psychological interpretation of the metaphorical target (i.e., the woman made the boy feel better); FLIT = a false rendering of the meaning of a metaphorical target (i.e., aspirin $\neq$ nurse); FPER = a false perceptual interpretation of a feature of the metaphorical target (i.e., the woman did not have red cheeks); and FPSY = a false psychological interpretation of the metaphorical target (i.e., the woman did not feel proud).

On the delayed test, children of both age levels were no longer able to suppress false recognition of gist-consistent distractors through verbatim mismatch. For 6-year-olds, *V* was not significantly greater than zero for TLIT or TPER distractors, though *V* was .07 for TPSY distractors. For 9-year-olds, *V* was not significantly greater than zero for any of the three types of gist-consistent distractors. There was little decline in gist processing at either age level. In 6-year-olds, the parameter *G* declined from an average of .10 to an average of .06, the decline being significant only for TPER distractors, $G^2(1) = 7.90, p < .01$. In 9-year-olds, a pattern similar to that in Kiernan's (1993) data was obtained: Between the immediate and delayed tests, the mean value of *G* nearly doubled. The increase was significant for TLIT distractors, $G^2(1) = 107.51, p < .01$, and for TPSY distractors, $G^2(1) = 26.46, p < .0001$. It

is important to note that the false-alarm rate for gist-consistent distractors (in the verbatim-acceptance condition) increased dramatically, doubling in 6-year-olds (from .17 to .34) and increasing sevenfold in 9-year-olds (from .07 to .49). Note that the false-alarm rate for gist-consistent distractors on the delayed test was actually higher for 9-year-olds than for 6-year-olds. Note, too, that across age levels, the mean proportion of correct rejections of gist-consistent distractors that were due to verbatim mismatches decreased from $.29 \div .88 = .33$ to $.01 \div .42 = .02$. Thus, 1 week after narrative presentation, the rejection of gist-consistent distractors had become largely a matter of passive nonacceptance that was due to lack of memory support.

Model validity results. Again, two types of findings bear on model validity: goodness of fit and parameter behavior. However, we confine ourselves to the first type of evidence, because the second is consistent with patterns already reported. Both qualitative and quantitative fit tests were again conducted. The relevant data for Reyna and Kiernan's (1995) immediate test (in which verbatim traces should still be accessible) are displayed in Figure 3 for both younger (Panel A) and older (Panel B) children. It can be seen that the prediction that the difference in acceptance rates between gist-consistent and gist-violating distractors should be greater in the gist-acceptance condition than in the verbatim-acceptance condition was confirmed for both age levels. With regard to quantitative tests, the fit statistic for the recollection-

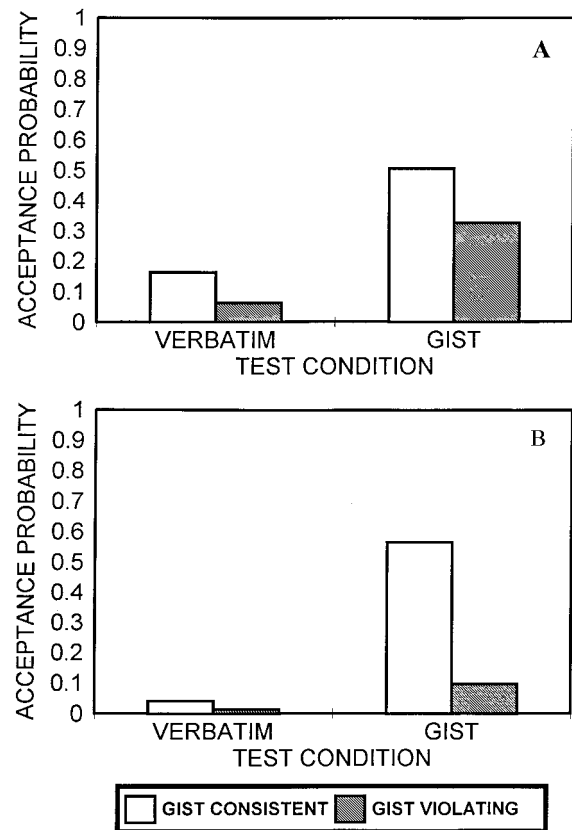


Figure 3. For Reyna and Kiernan's (1995) experiments, mean acceptance rates for gist-consistent and gist-violating distractors in the verbatim-acceptance and gist-acceptance conditions for 6-year-olds (Panel A) and 9-year-olds (Panel B).

rejection model that is given in the Appendix was computed for 12 data sets (3 distractor pairings $\times$ 2 age levels $\times$ 2 memory tests). Across the 12 data sets, the average computed value of the statistic was 2.95. The critical value that is required to reject the null hypothesis that the recollection rejection model fits the data is 5.99, which means that the model gave an excellent account of these data.

General Discussion

The objective of this research was to make progress on two fronts—namely, on the formulation of detailed process theories of developmentally plausible false-memory editing operations and on the evaluation of new methodological techniques that deliver direct measurements of amounts of false-memory editing. In the first subsection below, we summarize the progress that has been made on both fronts. In the second subsection, we sketch some implications of this work for child forensic interviewing. In the final subsection, we consider metamemorial approaches to false-memory editing.

Recollection Rejection: Theory, Method, and Data

Fuzzy-trace theory postulates an editing operation, recollection rejection, that is developmentally plausible because it is within the known cognitive capabilities of even young children. This operation neutralizes the feelings of familiarity that accompany false-but-gist-consistent events (e.g., eating a hot dog or drinking a Coke at a baseball game) by accessing verbatim traces of corresponding true events (“ate a hamburger,” “drank a 7-Up”) and generating mismatches at the level of verbatim detail. Such mismatches allow children to reject false-but-gist-consistent events, notwithstanding their familiarity. In effect, verbatim mismatches provide an explanation of how an event that is familiar could nevertheless not have been experienced (i.e., because it mismatches a very similar event that is known to have been experienced).

Certain findings from the memory development literature are broadly consistent with the notion that children spontaneously use recollection rejection to edit false-but-gist-consistent information out of their memory reports. For instance, because this operation depends on the retrieval of verbatim traces, the well-known tendency of such traces to become rapidly inaccessible over time implies that false reporting should increase as time passes, a finding that has often been reported for children and adults (Pipe et al., 1999; Reyna, 1992, 1995). Further, because recollection rejection relies on the retrieval of *verbatim* traces, memory-test cuing manipulations that selectively enhance the accessibility of surface information ought to decrease false reporting. This finding, too, has been reported (Brainerd & Mojardin, 1998; Brainerd, Reyna, & Kneer, 1995; Schacter, Israel, & Racine, 1999).

Such findings provided grounds for optimism that children use recollection rejection on standard memory tests, but more direct evidence was necessary. The production of such evidence requires a paradigm that is specifically designed to measure this editing operation. Release from recollection rejection is a paradigm with high face validity because it implements the definition of recollection rejection in a transparent manner: If recollection rejection sometimes suppresses acceptance of false-but-gist-consistent events, it follows that acceptance of such events will increase if the

normal constraint to accept only things that were actually experienced is removed, and if children are also allowed to accept things that are consistent with what was experienced (i.e., gist-consistent events). An increase in acceptance of false-but-gist-consistent events under such modified instructions will occur because gist-consistent events that are rejected owing to verbatim mismatches under standard instructions will now be accepted. Further, when children respond to memory tests under both types of instructions, fuzzy-trace theory’s conjoint recognition model can be used to measure amounts of recollection rejection at different age levels and in different treatment conditions.

To produce direct measurements of recollection rejection in children, we applied the release-from-recollection-rejection paradigm to data sets in which children responded to recognition tests for narrative statements in both standard (verbatim-acceptance) conditions and modified (gist-acceptance) conditions. These analyses produced several instructive findings about recollection rejection, but the following five are perhaps of greatest import for future research on false-memory editing.

First, there were dramatic developmental improvements in recollection rejection regardless of whether “development” was defined as between-age comparisons or as comparisons of developmentally delayed children with control children. Developmental comparisons revealed a doubling in recollection rejection in two of the data sets and a fivefold increase in the third data set. Those developmental increases were sufficiently large to overcome parallel increases in gist processing, thereby yielding net developmental reductions in false-memory reports *even though the memorial basis for such reports was increasing*. Also, owing to increases in recollection rejection, children’s basis for rejecting false-but-gist-consistent events changed fundamentally with development. At earlier developmental levels, correct rejection of such events was chiefly a matter of passive nonacceptance owing to inadequate memory support from gist processing. At later developmental levels, correct rejection of such events became memory-based rejection that occurred despite memory support for acceptance from gist processing.

Second, despite large developmental improvements in recollection rejection, there was clear evidence that this operation suppressed false acceptance of gist-consistent events at the earliest developmental level in each data set. Across the data sets, the average probability that children at the earliest developmental level used recollection rejection to suppress acceptance of false-but-gist-consistent statements was .16. Therefore, children at still earlier developmental levels (i.e., kindergartners and preschoolers) are presumably capable of using this operation, and the demonstration of this capability becomes an important objective of future research. Third, both qualitative fit tests (Figures 1, 2, and 3) and quantitative fit tests (Equation A11) showed that the recollection-rejection model gave a good account of the three data sets. From the standpoint of future research, this is an especially probative result because it confirms that the release-from-recollection-rejection paradigm is an appropriate tool for studying false-memory editing in children.

Fourth, in all three data sets, parameter validity tests revealed that *V* and *G*—the parameters that measure verbatim-based rejection and gist-based acceptance of distractors, respectively—reacted to experimental manipulations in ways that are congruent with their theoretical definitions. For example, consistent with the notion that *V* measures retrieval of labile verbatim traces and *G*

measures retrieval of more stable gist traces, V always decreased more than G , as time passed.⁴ These parameter-validity results provide further confirmation that the release-from-recollection-rejection paradigm is an appropriate tool for studying false-memory editing in children. Fifth and finally, the parameter V displayed more developmental variability than the parameter G . The interesting conclusion is that although children's ability to process the gist of experience improves during the elementary school age range, their ability to process the exact verbatim content of experience improves more. This is a surprising conclusion from the perspective of traditional theories of memory development, which place principal emphasis on the growth of semantic capabilities (Reyna & Kiernan, 1994). It is not surprising from the perspective of fuzzy-trace theory, which stresses parallel developmental trends in verbatim and gist memory (Brainerd & Reyna, 1998a; Reyna & Brainerd, 1995) and which distinguishes developmental trends in *memory* from developmental trends in reliance on memory in *reasoning* (Reyna, 1992).

Exploiting Recollection Rejection in Child Forensic Interviewing

Concerns about the accuracy of children's memory reports in forensic interviews supplied the original impetus for the study of children's false memories. One result of such work, which was mentioned earlier, is the construction of model child interviewing protocols that restrict the suggestive forms of questioning that have been found to create false memories (e.g., Lamb et al., 1999, 2000; Poole & Lamb, 1998). When it comes to controlling deliberate or inadvertent memory falsification, these protocols are a major advance over traditional interviewing methods. However, factors that induce false memories cannot be entirely eliminated in forensic contexts, and further tools are needed to deal with this fact. How could the recollection-rejection operation and the release-from-recollection-rejection paradigm be exploited for this purpose? Two general possibilities suggest themselves.

First, the process theory of recollection rejection provides explanatory leverage on current interviewing procedures, thereby allowing us to predict new procedures that will be effective in reducing false reports. Regarding explanation, the process theory supplies a unified account of why many recommended procedures, which were adopted on a variety of empirical and practical grounds, are successful in reducing false reports. Five examples of such recommended procedures (e.g., Lamb et al., 1999; Poole & Lamb, 1998) are that (a) interviews should occur as soon as possible after events, (b) interviews should be preceded by a settling-in period in which the types of responses that an interviewer is seeking are clearly explained, (c) settling-in instructions should stress that responses in the upcoming interview must be confined to events that are clearly remembered as having happened, (d) children should be pretrained with concrete examples of true events that should be reported and related false events that should not be reported, and (e) interviewers should conclude with opportunities for children to review and clarify answers. It is easy to see that each of these procedures should increase recollection rejection by enhancing one or more of the components of its process theory—that is, by enhancing the availability of verbatim traces (procedure a), by enhancing the ability of false-but-gist-consistent events to cue the retrieval of verbatim traces of true events (procedures c and d), and by enhancing the perception that

verbatim mismatches with true events should override the familiarity of false-but-gist-consistent events (procedures b, c, d, and e).

Beyond explaining the success of existing interviewing procedures, the specificity of the process theory supplies the more important advantage of allowing investigators to predict, in advance of field studies, new procedures that will be especially likely to reduce false reports. The basic predictive rule is simple: A new procedure is likely to be successful if it transparently enhances one of the components of the process theory *without increasing children's tendency to rely on gist memories*. A corollary is that new procedures that enhance more than one component of the process theory are especially likely to be successful.

A second way to exploit recollection rejection in forensic contexts is to use the release-from-recollection-rejection paradigm, under either laboratory or field conditions, (a) to pinpoint the actual memory effects of forensic interviewing procedures and (b) to measure the actual memorial bases for individual differences in false reporting. Concerning (a), the release-from-recollection-rejection paradigm could be used to determine whether specific interviewing procedures (new or existing) affect false reporting by influencing verbatim processing, gist processing, response bias, or some combination of these factors. The paradigm is just as applicable to the event memories that figure in controlled studies of forensic interviewing, and in field interviews, as it is to memories of lists of words, pictures, or sentences. Applicability does not turn on the nature of the to-be-remembered material, nor does it turn on whether memory reports are obtained under laboratory or field conditions. Instead, applicability turns on the presence of three types of memory questions in interviews (for true events, false-but-gist-consistent events, and gist-violating events) and on obtaining responses to such questions under both verbatim- and gist-acceptance instructions.

Concerning point (b), a property of the release-from-recollection-rejection paradigm that could be very useful in forensic work is that the parameters V , G , and β can be estimated for individual children as well as for groups. To obtain individualized estimates, it is only necessary that a child respond to memory questions under both verbatim- and gist-acceptance instructions (and that there be sufficient questions for estimates to be reliable). In principle, individualized parameter estimates could be obtained for any type of material, but one can imagine the construction of standardized forensic instruments that present different sets of to-be-remembered material to children, with questions about some sets being posed under verbatim-acceptance instructions and questions about other sets being posed under gist-acceptance conditions. The resulting parameter estimates might then be exploited in

⁴ It might be thought that the use of the release-from-recollection-rejection paradigm in applied contexts that involve substantial delays in memory testing is undercut by the fact that recollection rejection dropped to near-floor levels after 1 week. In the data sets that we analyzed, however, the experimental conditions were not such as would be expected to create good long-term verbatim memories: Children were exposed to several narratives at once, none of which was particularly distinctive and none of which was repeated. Other research shows much better long-term verbatim memory when children receive multiple exposures to target material (e.g., Brainerd, Reyna, Howe, & Kingma, 1990) or when they have experienced distinctive, emotionally arousing events of the sort that figure in forensic interviews and sworn testimony (e.g., Howe, 1997; Pipe et al., 1999).

at least two ways. First, they might be used diagnostically to identify children who may be especially prone to false reporting (owing to low V values, high G values, or high β values). Second, individualized estimates might be used to address the perennial question of predicting individual differences in false reporting (Reyna, Holliday, & Marche, in press). Given the importance of false reports in forensic contexts, there has been considerable interest in determining whether there are predictor variables (e.g., IQ, personality factors) that correlate strongly with false reporting (for a review, see Bruck, Ceci, & Melnyk, 1997). Individualized parameter estimates provide a more precise approach to this question because raw false reporting (the usual criterion variable in individual-differences studies) can be replaced by estimates of the processes that are actually responsible for false reports. Because the evidence suggests that the processes that V , G , and β measure are dissociated, it is to be expected that predictor variables that correlate strongly with one parameter will not necessarily correlate strongly with the others (Reyna & Brainerd, 1998). This, in turn, could explain why individual-differences research has not yet been successful in identifying predictor variables that correlate strongly with raw false reporting.

Metacognitive False-Memory Editing

In closing, another form of false-memory editing has been investigated in adults, which we call *metamemorial suppression* because it involves the application of global strategies (i.e., strategies that are not specific to particular items but that apply list-wide to studied material) that assess the types of events that are most apt to have been experienced. Strack and Bless (1994) proposed that various manipulations may induce participants to require stronger across-the-board memorial confirmation before they are willing to report events as having been experienced. For instance, participants might be warned of the presence of numerous false items that will be mistakenly accepted if they are not careful, or they might be informed that penalties will be imposed for false reports. Such instructions could induce deliberate self-monitoring strategies in which participants refuse to report any event as true unless memorial confirmation is overwhelming. Alternatively, prior to memory tests, certain types of target experiences (e.g., inspecting color photographs) may cause participants to believe that their later recollections will be accompanied by vivid perceptual phenomenologies, leading them to follow a strategy of refusing to report any memory as true that is not accompanied by such phenomenologies (e.g., Schacter et al., 1999).

Koriat and Goldsmith (1994, 1996), in particular, have devised a memory self-monitoring paradigm that makes use of a method called optional reporting, in which participants respond to a standard memory test or to one in which they have the choice of withholding or reporting remembered items of information. Consistent with the notion of metamemorial suppression, accuracy levels (e.g., total true information recalled divided by total information recalled) are normally higher in optional-report conditions. Further, participants in such conditions have reported that they deliberately screen candidate items of information, as they come to mind, weeding out those that are believed to be false in order to enhance the accuracy of memory performance.

Metamemorial suppression may be important in adults, but there are at least two reasons to conclude that it does not tell the full story of false-memory editing. First, traditional definitions of

metamemorial processes (e.g., Bjorklund, 1999) emphasize conscious control of performance via the deliberate application of mnemonic strategies. However, much or most adult false-memory editing may occur automatically, without comprehension, and even outside conscious awareness. For example, although adults spontaneously use information about memory strength to delay the output of certain items during free recall, they are typically unaware that they are doing so (Brainerd, Reyna, Howe, & Kevershan, 1990). Second and more important from a developmental perspective, there is a large literature on metamemory in children (for a review, see Schneider & Bjorklund, 1998), a key conclusion of which is that metamemorial processes such as those that have been studied in adults are normally not present until late childhood and early adolescence. Hence, the hypothesis that would surely be favored by most developmental researchers is that the study of simpler, more automatic operations, such as recollection rejection, is a more productive approach to false-memory editing during the preschool and elementary school years.

Nevertheless, Koriat et al. (2001) reported some studies that they interpreted as showing that children as young as age 7 deliberately screen out false information. Specifically, these investigators used the optional-report procedure to study children's memory for a picture story and found that children's reports, like adults', were more accurate in an optional-report condition. Although intentional screening may therefore be present early in development, two points should be noted concerning the relation between intentional screening and recollection rejection. First, there is no incompatibility between the two processes, at a theoretical level, and indeed available evidence suggests that adults rely on both types of false-memory editing (for a review, see Brainerd, Reyna, Wright, & Mojardin, 2001). Second, as yet, neither the theoretical nor the empirical case for intentional screening in young children is as compelling as the theoretical and empirical case for recollection rejection. On the theoretical side, recollection rejection is developmentally plausible (because it involves memory mechanisms that are within the known capabilities of young children), and it has been found to respond to experimental manipulations in accordance with theoretical predictions. On the empirical side, we have reported several findings that are consistent with the view that even young children are able to reject false-but-gist-consistent events by retrieving verbatim traces of the corresponding true events.

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(Appendix follows)

Appendix

Recollection Rejection Model

A goodness-of-fit test is obtained for the recollection-rejection model by defining the model over the joint probabilities of accepting and rejecting gist-consistent and gist-violating distractors in two instructional conditions. For the verbatim-acceptance condition, the model's expression for the probabilities of accepting both a gist-consistent distractor and its corresponding gist-violating distractor is

$$p_{VA}(A_T A_F) = [(1 - V)G + (1 - V)(1 - G)\beta_{VA}][(1 - V)(1 - G)\beta_{VA}]. \quad (A1)$$

The expression for accepting the gist-consistent distractor and rejecting the gist-violating distractor is

$$p_{VA}(A_T R_F) = [(1 - V)G + (1 - V)(1 - G)\beta_{VA}] \times [V + (1 - V)G + (1 - V)(1 - G)(1 - \beta_{VA})]. \quad (A2)$$

The expression for rejecting the gist-consistent distractor and accepting the gist-violating distractor is

$$p_{VA}(R_T A_F) = [V + (1 - V)(1 - G)(1 - \beta_{VA})][(1 - V)(1 - G)\beta_{VA}]. \quad (A3)$$

The expression for rejecting both the gist-consistent distractor and the gist-violating distractor is

$$p_{VA}(R_T R_F) = [V + (1 - V)(1 - G)(1 - \beta_{VA})] \times [V + (1 - V)G + (1 - V)(1 - G)(1 - \beta_{VA})]. \quad (A4)$$

For the gist-acceptance condition, the model's expression for the probabilities of accepting both a gist-consistent distractor and its corresponding gist-violating distractor is

$$p_{GA}(A_T A_F) = [V + (1 - V)G + (1 - V)(1 - G)\beta_{GA}][(1 - V)(1 - G)\beta_{GA}]. \quad (A5)$$

The expression for accepting the gist-consistent distractor and rejecting the gist-violating distractor is

$$p_{GA}(A_T R_F) = [V + (1 - V)G + (1 - V)(1 - G)\beta_{GA}] \times [V + (1 - V)G + (1 - V)(1 - G)(1 - \beta_{GA})]. \quad (A6)$$

The expression for rejecting the gist-consistent distractor and accepting the gist-violating distractor is

$$p_{GA}(R_T A_F) = [(1 - V)(1 - G)(1 - \beta_{GA})][(1 - V)(1 - G)\beta_{GA}]. \quad (A7)$$

The expression for rejecting both the gist-consistent distractor and the gist-violating distractor is

$$p_{GA}(R_T R_F) = [(1 - V)(1 - G)(1 - \beta_{GA})] \times [V + (1 - V)G + (1 - V)(1 - G)(1 - \beta_{GA})]. \quad (A8)$$

The standard Fisher-type likelihood function that expresses the a posteriori probability of the data, given the observable probabilities on the left sides of Equations A1 through A8, is

$$L_6 = [p_{VA}(A_T A_F)]^{N_{VA}(A_T A_F)} \times [p_{VA}(A_T R_F)]^{N_{VA}(A_T R_F)} \times [p_{VA}(R_T A_F)]^{N_{VA}(R_T A_F)} \times [p_{VA}(R_T R_F)]^{N_{VA}(R_T R_F)} \times [p_{GA}(A_T A_F)]^{N_{GA}(A_T A_F)} \times [p_{GA}(A_T R_F)]^{N_{GA}(A_T R_F)} \times [p_{GA}(R_T A_F)]^{N_{GA}(R_T A_F)} \times [p_{GA}(R_T R_F)]^{N_{GA}(R_T R_F)}, \quad (A9)$$

where the exponent of each term is the number of times that the indicated joint event occurred in the data. This likelihood function has six degrees of freedom (i.e., there are a total of six empirical probabilities that are free to vary). The likelihood function for the recollection-rejection model is

$$L_4 = f[p_{VA}(A_T A_F)]^{N_{VA}(A_T A_F)} \times f[p_{VA}(A_T R_F)]^{N_{VA}(A_T R_F)} \times f[p_{VA}(R_T A_F)]^{N_{VA}(R_T A_F)} \times f[p_{VA}(R_T R_F)]^{N_{VA}(R_T R_F)} \times f[p_{GA}(A_T A_F)]^{N_{GA}(A_T A_F)} \times f[p_{GA}(A_T R_F)]^{N_{GA}(A_T R_F)} \times f[p_{GA}(R_T A_F)]^{N_{GA}(R_T A_F)} \times f[p_{GA}(R_T R_F)]^{N_{GA}(R_T R_F)}, \quad (A10)$$

where the f that precedes each term indicates that the empirical probabilities inside the brackets have been replaced by the corresponding theoretical expressions in Equations A1 through A8. This second likelihood function has four degrees of freedom because only the parameters V , G , β_{VA} , and β_{GA} are free to vary. Therefore, the goodness-of-fit test for the recollection rejection model is

$$G^2(2) = -2\ln[L_4 \div L_6]. \quad (A11)$$

For large N , the statistic G^2 is distributed approximately as chi-square, which means that critical values of the chi-square distribution are used in significance testing. Thus, to test the null hypothesis that the recollection-rejection model fits some set of sample data, the critical value of $G^2(2)$ is 5.99 if the confidence level is set at .05.

When the recollection-rejection model is found to fit data from the conditions of some experiment (i.e., Equation A11 does not yield null hypothesis rejections), the parameters V , G , β_{VA} , and β_{GA} can be estimated and interpreted. Estimated values of these parameters will normally be subjected to two types of significance tests—namely, within condition and between conditions. A within-condition test evaluates the null hypothesis that two parameters have the same value in a given condition (or conditions) of the experiment (e.g., $\beta_{VA} = \beta_{GA}$). A between-conditions test evaluates the null hypothesis that a given parameter (e.g., V) has the same value in different conditions of the experiment.

Within-condition tests are conducted as follows. Let L_4 be the likelihood of the data, under Equation A10, in which all four parameters of the model are free to vary, and let L_3 be the likelihood of the same data under the constraint that two of the parameters must have the same value (e.g., $\beta_{VA} = \beta_{GA}$). For large N , the test statistic

$$G^2(1) = -2\ln(L_3 \div L_4), \quad (A12)$$

is distributed approximately as chi-square, with a critical value of 3.86 when the confidence level is set at .05. Between-conditions tests are conducted as follows. Let $L_{4,A}$ be the likelihood of the data, under Equation A10, for some Condition A of an experiment, and let $L_{4,B}$ be the likelihood of the data, under Equation A10, for some Condition B of an experiment. Then, $[L_{4,A} \times L_{4,B}]$ is the joint likelihood of the data when all eight parameters are free to vary (2 conditions $\times$ 4 parameters), and $[L'_{4,A} \times L'_{4,B}]$ is the likelihood of the same data under the constraint that one of the model's four parameters must have the same value in both conditions (e.g., $V_A = V_B$). For large N , the test statistic

$$G^2(1) = -2\ln([L'_{4,A} \times L'_{4,B}] \div [L_{4,A} \times L_{4,B}]) \quad (A13)$$

is distributed approximately as chi-square, with a critical value of 3.86 when the confidence level is set at .05.

Received April 10, 2001
Revision received August 20, 2001
Accepted August 20, 2001 ■